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RESEARCH ARTICLE

Control of Movement

Acute cross-education effect on force production and central/peripheral responses to unilateral eccentric and concentric resistance exercise in elbow flexors

Omar Valdés,[1,4] Carlos Rehbein,[1,2] Oscar Núñez,[1] Emeric Chalchat,[3,4,5] Julien Siracusa,[3,4,5] Sebastián García-Vicencio,[5,6] Claire Thomas-Junius,[5] Mounir Chennaoui,[5] Vincent Martin,[4,7] and Luis Peñailillo[1]

[1]*Exercise and Rehabilitation Sciences Institute, School of Physical Therapy, Faculty of Rehabilitation Sciences, Universidad Andres Bello, Santiago, Chile;* [2]*Sport Optimization & Rehabilitation Laboratory, University of Connecticut, Storrs, Connecticut, United States;* [3]*Institut de Recherche Biomédicale des Armées, Bretigny-Sur-Orge, France;* [4]*AME2P, Université Clermont Auvergne, Clermont-Ferrand, France;* [5]*LBEPS, Univ Evry, IRBA, Université Paris Saclay, Evry, France;* [6]*HumanMotion Analysis, Humanfab, Aix-en-Provence, France; and* [7]*Institut Universitaire de France (IUF), Paris, France*

Abstract

The cross-education effect (CE) is the transference of neuromuscular adaptations from a single exercised limb to the contralateral nonexercised limb, which seems to differ between exercise modalities. We compared the acute CE of unilateral eccentric (ECC) and concentric (CONC) resistance exercises on neuromuscular function and force production changes of the nonexercised elbow flexors (EF). Healthy men were randomly allocated into ECC ($n = 15$) or CONC ($n = 15$) groups. The effects of control ($CTRL_{CONC}$ or $CTRL_{ECC}$: 30 min of sitting) and exercise conditions (5 sets × 10 repetitions at 80% of either ECC or CONC 1-repetition maximum) of the dominant EF were measured. Maximal voluntary isometric contraction (MVIC), rate of torque development (RTD), corticospinal excitability (CSE), voluntary activation (VA), and peripheral factors were measured before and immediately after $CTRL_{CONC}$/$CTRL_{ECC}$ or CONC/ECC. Surface electromyography amplitude integral ($sEMG_i$) from the biceps brachii (BB) muscle was monitored during exercise. Physical-mental demands (NASA-TLX) were assessed after exercise. ECC performed 26.2% greater exercise volume than CONC ($P = 0.01$). ECC showed lesser BB $sEMG_i$ ($P = 0.04$) than CONC in the exercised EF; however, it induced threefold greater irradiated $sEMG_i$ to the nonexercised BB than CONC ($P = 0.04$) during exercise. NASA-TLX was unchanged ($P = 0.81$). The nonexercised EF maintained MVIC, VA, cortical silent period, and peripheral factors after ECC and CONC ($P > 0.05$), but lower RTD was observed after ECC (23.5%–29.4%; $P < 0.05$) and CONC (10.8%; $P < 0.05$). Lower CSE at 75% of MVIC occurred after ECC (−17%; $P = 0.02$) compared with $CTRL_{ECC}$, which was correlated to RTD decreases ($r = 0.45$–0.56, $P = 0.02$–0.04). ECC induces acute contralateral reductions in CSE that are associated with decreases in RTD.

NEW & NOTEWORTHY Unilateral eccentric and concentric resistance exercises can modulate the neuromuscular response differently, which influences the transference of the force production capacity to the contralateral nonexercised muscle. Unilateral eccentric exercise induced decreases in corticospinal excitability with associated reductions in the early phase of the rate of torque development, which differs from the changes observed after concentric exercise. Thus, both exercise modalities seem to change the excitability differently.

contralateral effect; cross-education; immobilization; lengthening contractions; rate of torque/force development

Correspondence: L. Peñailillo (luis.penailillo@unab.cl).
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INTRODUCTION

Unilateral resistance training (URT) can increase muscle force production of the nontrained contralateral homologous muscle, a phenomenon known as the "cross-education effect" (CE) (1–3). The CE induced by URT could be used in clinical settings, for example, to spare muscle force production during unilateral limb immobilization (4), rehabilitation of unilateral musculoskeletal injuries (5), and even in neurological conditions in which the direct resistance exercise of the one limb is contraindicated for injury protection or a limb unable to perform moderate-high intensity resistance exercise directly. Dynamic URT (i.e., eccentric and concentric contractions) has been shown to induce greater CE on muscle strength than isometric training (6). Eccentric training involves active lengthening of the muscle under tension, whereas concentric training involves active shortening of the muscle under tension (7). Each training mode has distinct physiological responses and neuromuscular adaptations (7, 8), but evidence has shown greater force production increases in the contralateral nontrained muscle after eccentric compared with concentric URT protocols (4, 9–12). Even though the exact neuromuscular mechanisms underlying the superior effect of eccentric URT to increase force production in the nontrained limb have not been fully identified, our current knowledge suggests that neural changes confined at supraspinal and spinal levels could be involved in this effect (1, 3).

The increases in force production following short-term URT are a cumulative result of multiple single exercise sessions, which induced changes at supraspinal and spinal levels (1, 13, 14). However, the acute CE on force production induced by unilateral eccentric compared with concentric resistance exercise has been less studied than the short-term training effects (4, 9, 11, 12, 15). Regarding the acute neuromuscular changes in the nontrained limb induced by unilateral resistance exercise, controversial results exist concerning force production and corticospinal excitability. For example, Miller et al. (16) found acute contralateral increases in maximal voluntary isometric contraction (MVIC) following unilateral concentric resistance exercise, whereas unilateral eccentric resistance exercise induced contralateral reductions in MVIC. In another study (17), no changes in concentric maximal torque were observed in the nonexercised muscles after unilateral concentric protocol, whereas they reported an 11% increase in the eccentric maximal torque of nonexercised knee extensors after contralateral eccentric exercise of the homologous muscles (17). Furthermore, the corticospinal excitability (CSE) of nonexercised biceps brachii (BB) has reported increases (15, 18), decreases (19, 20), and no changes (15) immediately after unilateral exercise. For instance, van der Groen et al. (15) found increases and no changes in CSE assessed by resting motor threshold in the nonexercised elbow flexors after unilateral concentric and eccentric resistance exercise, respectively. The changes at peripheral levels (i.e., electrically evoked torque and maximal compound wave) also have been reported with different results in the nonexercised contralateral muscles after unilateral resistance exercise (21, 22). For example, electrically evoked doublet torque decreases were reported in the nonexercised knee flexors after contralateral submaximal fatiguing eccentric contractions (22). However, no changes in electrically evoked twitch were reported in the nonexercised plantar flexors after unilateral damaging eccentric contractions (21). Investigating the acute responses to different exercise modalities may be important to understand the contralateral adaptative effects at central and peripheral levels of the neuromuscular system in the nonexercised muscles due to CE. However, previous studies assessing the contralateral responses have focused on changes in the maximal strength/torque, whereas the rate of force/torque development (RDF/RTD) has been less explored (8, 23). Yet, the RFD/RTD could represent a more sensible marker of changes in the neuromuscular system, as previously pointed out (24). In particular, early phases of RFD/RTD are thought to reflect central neural drive, whereas later phases may be more influenced by peripheral factors such as muscle-tendon stiffness and cross-bridge kinetics within skeletal muscle (23, 24). Therefore, analyzing RFD/RTD changes across different time windows may help elucidate whether acute neuromuscular alterations occur at central or peripheral levels. Furthermore, the physical and mental demands imposed by eccentric and concentric URT could also influence force production performance transference to the nonexercised limb (1, 2) due to specific task complexity imposed by these contractions (25). Nevertheless, the physical and mental demands of acute bouts of eccentric and concentric resistance exercise have not been associated previously with CE in force production.

Therefore, this study aimed to compare the unilateral acute changes in force production, CSE, voluntary activation (VA) level, and peripheral changes in the nonexercised elbow flexors immediately after an acute eccentric (ECC) or concentric (CONC) unilateral resistance exercise bout. In addition, we examined the acute physical and mental demands after exercise to explore its possible associations with the CE on force production. Compared with unilateral concentric resistance exercise, we hypothesized that unilateral eccentric resistance exercise would result in greater acute associated decreases in force production, CSE, and VA of the nonexercised elbow flexors.

METHODS

Participants

Thirty healthy men (24.70 ± 3.8 yr old, 78.51 ± 6.6 kg, 174.63 ± 6.8 cm) recruited from the university community participated in this study. Only men were recruited in the present study to avoid possible differences between men and women in response to acute neuromuscular fatigue (26). The participants were either right ($n = 23$) or left ($n = 7$) handed dominants, as assessed by the Edinburgh Handedness Inventory questionnaire (27). The present study excluded participants who had been performing resistance training regularly (2–3 times/wk during the last year) and/or had an upper limb injury in the previous year (4). According to reported changes in CSE (η_p^2 of 0.36) in the nonexercised muscle induced by unilateral concentric and eccentric resistance exercise of a previous study (15), we estimated a sample size of $n = 12$, considering an α level of 0.05 and 0.80 for statistic power one-way analysis of variance (ANOVA) in G*Power (G*Power 3.1, Germany). However, considering a potential 20% drop-off, 15 participants per group were recruited. All participants were

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instructed to refrain from upper body resistance training, any nutritional supplement, or anti-inflammatory medication, and maintain their normal daily routines during the experimental period. In addition, participants were asked to refrain from drinking caffeinated drinks or alcohol for at least 6 h before testing sessions. Written informed consent was obtained from all participants, and the study was approved by the Institutional Ethics Committee (No. 002/2022) and conducted according to the Declaration of Helsinki.

Experimental Design

All participants were evaluated in the laboratory three times at the same time of day (±1 h). A schematic representation of the experimental design is shown in Fig. 1A. In the first visit, all participants performed a familiarization session, in which anthropometric and neuromuscular testing were performed. The neuromuscular measurements began with determining the optimal and supramaximal intensity of peripheral nerve stimulation (PNS) at rest (described in *Peripheral Nerve Stimulation*). Following a brief warm-up of the elbow flexors, single twitch (Tw), doublet of low frequency (Db10), and high frequency (Db100) were evoked with PNS, respectively. After PNS at rest, the participants performed the first maximal voluntary isometric contraction (MVIC-1) of the elbow flexors. Subsequently, the percentage of VA was determined with the twitch interpolation technique using PNS (%VA PNS) (described in *Peripheral Nerve Stimulation*). Thereafter, the hotspot and recruitment curve to determine the optimal and supramaximal intensity with transcranial magnetic stimulation (TMS) for BB was performed (described in *Transcranial Magnetic Stimulation*). After the recruitment curve, the percentage of VA was determined using TMS (%VA TMS), as described in *Transcranial Magnetic Stimulation*. Finally, a second MVIC (MVIC-2) of the elbow flexors was performed

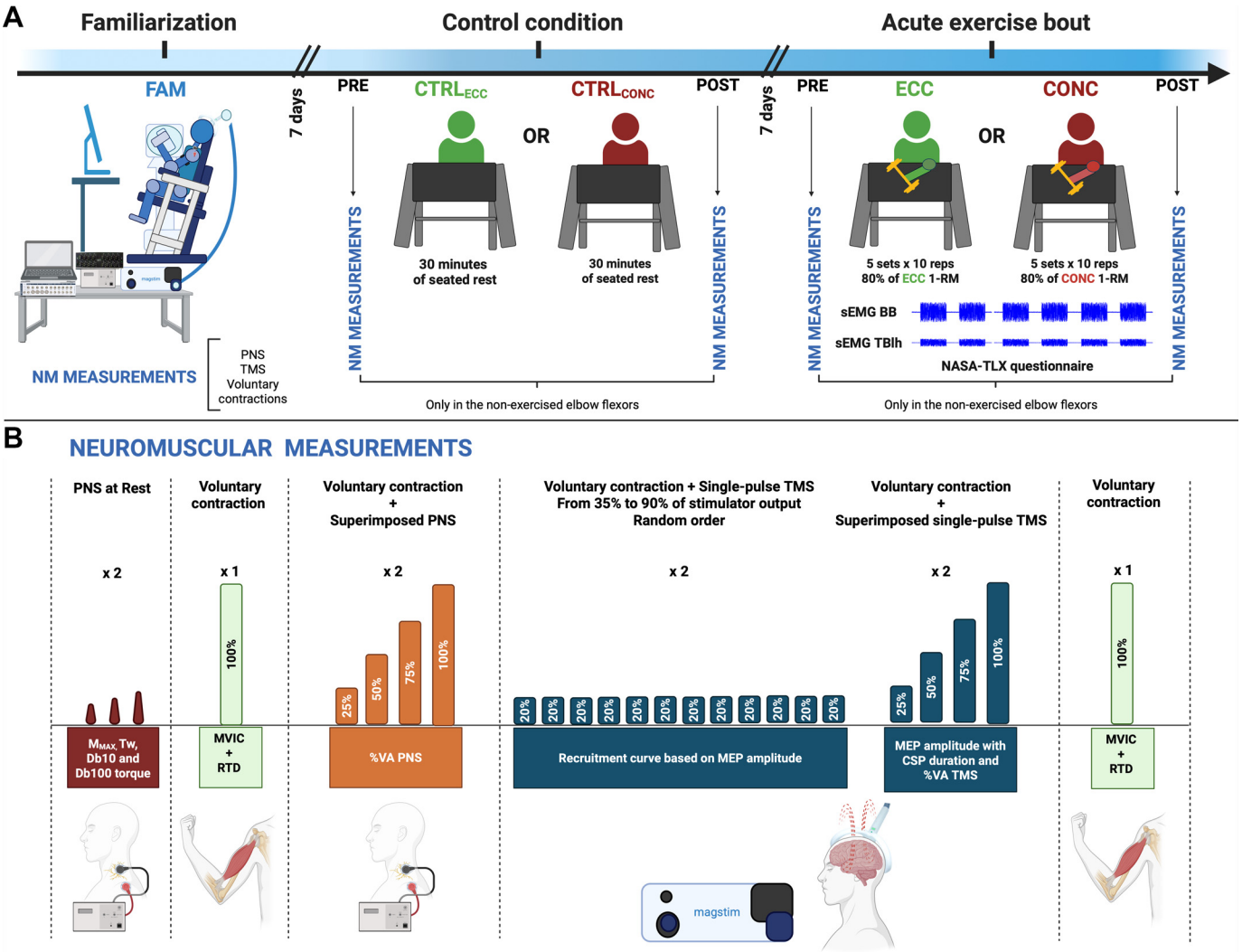

Figure 1. Representative overview of the study design (*A*) and neuromuscular testing measurements (*B*). NM, neuromuscular; FAM, familiarization; PNS, peripheral nerve stimulation; TMS, transcranial magnetic stimulation; $CTRL_{ECC}$, control condition of eccentric; $CTRL_{CONC}$, control condition of concentric; ECC, eccentric; CONC, concentric; sEMG, surface electromyography; BB, biceps brachii; TBlh, triceps brachii long head; NASA TLX, NASA task load index; 1-RM, one repetition maximum; Rep, repetitions; MVIC, maximal voluntary isometric contraction; RTD, rate of torque development; M_{MAX}, maximal compound wave; Tw, single twitch; Db10, doublet 10-Hz stimuli; Db100, doublet 100-Hz stimuli; %VA PNS, percentage of voluntary activation by peripheral nerve stimulation; MEP, motor-evoked potential; CSP, cortical silent period; %VA TMS, percentage of voluntary activation by transcranial magnetic stimulation.

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without any stimulation to check the neuromuscular fatigue induced by the testing protocol (Fig. 1*B*). Based on the familiarization assessment, the participants were allocated to a concentric (CONC; $n = 15$, 12 right- and 3 left-handed dominant) or eccentric (ECC; $n = 15$, 11 right- and 4 left-handed dominant) group. The allocation was random through a sequence of numbers generated by a computer that considered hand dominance and MVIC to obtain a similar distribution between groups. One week after the familiarization session, participants developed a control condition ($CTRL_{CONC}$ and $CTRL_{ECC}$) in which the neuromuscular measurements were taken before and immediately after 30 min of seated rest (Fig. 1*A*). Finally, 1 wk after $CTRL_{CONC}$ and $CTRL_{ECC}$, participants developed a single CONC or ECC unilateral resistance exercise bout with the dominant elbow flexors (described below) in which the neuromuscular measurements were taken before and immediately after exercise in the same manner as described earlier. During the CONC and ECC exercise sessions, bilateral recordings of surface electromyography (sEMG) from biceps brachii (BB) and triceps brachii long head (TBlh) were taken. Immediately after CONC and ECC exercise, MVIC and sEMG activity were measured from both arms, and NASA task load index (NASA-TLX) was completed by participants to assess the mental-physical demand of the exercise. All neuromuscular measurements were taken from the nonexercised arm and were performed in two attempts during $CTRL_{CONC}$, $CTRL_{ECC}$, CONC, and ECC conditions.

One-Repetition Maximum Testing and Exercise Protocols

A warm-up consisting of five repetitions at a self-estimated 50% of the maximum was performed before the one-repetition maximum (1-RM) test. To assess the concentric and eccentric 1-RM strength of the elbow flexors, adjustable dumbbells (0.5 kg) on the preacher curl bench (FFITTECH, Viseu, Portugal) were used based on previous recommendations (4, 15). For the $1\text{-}RM_{CON}$, the dumbbell was lifted by participants from the starting position with the elbow joint fully extended (0° of elbow flexion) to maximal elbow flexion (135° of elbow flexion) in 4 s (4). The $1\text{-}RM_{ECC}$ was determined by requesting the participants to descend a dumbbell from a fully flexed (135° of elbow flexion) elbow to a full elbow extension (0° of elbow flexion) in 4 s (15). 1-RM values were obtained between 3 and 5 attempts.

Both groups used adjustable dumbbells to perform a resistance exercise bout of the dominant elbow flexor muscles. ECC and CONC consisted of five sets of 10 repetitions at 80% of the $1\text{-}RM_{ECC}$ or $1\text{-}RM_{CONC}$, respectively. Contraction time was controlled using a metronome (Cifra Club for iOS, Minas Gerais, Brazil) to achieve 4 s of execution during each repetition of a full range of movement (ROM) (4). The CONC exercise consisted of a 4-s concentric contraction from 0° of elbow flexion to 135° of elbow flexion (total ROM = 135°), and the ECC exercise consisted of a 4-s eccentric contraction from full elbow flexion (135°) to 0° of elbow flexion (total ROM = 135°) (4), while the investigator lifted the dumbbell back to the starting position for subsequent repetitions. A 2-s resting time between repetitions and a 3-min resting time between sets were provided for both exercise groups.

Neuromuscular Setup

Participants were seated on an isokinetic dynamometer (ISOforce, TUR, Germany) using custom-built elbow support to maintain the shoulder and elbow joints at 45° and 90° of flexion, respectively (Fig. 1*A*). The forearm was secured to the dynamometer to maintain a neutral wrist position (0° of flexion) with the rotation axis of the dynamometer aligned ~1 cm above the lateral epicondyle. Single differential surface Ag (10-mm length with 10-mm interelectrode distance) electrodes (DE-2.1 EMG sensor Delsys Inc., Boston, MA) were used to record the sEMG amplitude from BB and TBlh (EMG; Bagnoli-16 Delsys Inc., Boston, MA). The sEMG was recorded at a sampling frequency of 2,000 Hz, using a gain of 1,000, and a bandpass filter of 20–500 Hz. A reference electrode was placed in the wrist over the dorsal surface of the hand. The skin was prepared by removing any hair and death cells and cleaned with 70% isopropyl alcohol swabs before the placement of electrodes. For BB and TBlh, the electrode placement was based on previous recommendations (4, 15, 28). Torque and sEMG signals were amplified and recorded using a PowerLab acquisition system (PowerLab 16/35, ADInstruments, Australia) synchronized with LabChart software (LabChart 8.1, ADInstruments, Australia) to perform the signal analyses. The PNS stimulus was delivered using a constant electrical stimulator (DS7A, Digitimer, UK) over the clavicle (Erb's point) (28) and TMS stimulation was delivered using a Magstim 200^2 unit (Magstim Co. Ltd, Whitland, UK). During the experiment, visual feedback of torque (without stimulation output feedback) was displayed on a monitor in front of the participants during all sessions (Fig. 1*A*).

Elbow Flexors MVIC and RTD

Each participant performed 3-s warm-up contractions at 30%, 50%, 70%, 80%, and 90% at the self-perceived MVIC with a 45-s rest between contractions before MVIC-1 (4). Strong verbal encouragement and visual feedback were provided to all participants, who were instructed to "pull as hard and fast as possible" (4). The highest peak torque (N·m) from each measurement was used for further analysis. RTD was measured as the slope of MVIC torque trace of each MVIC at the time windows of 0–50 ms (RTD_{0-50}), 0–100 ms (RTD_{0-100}), 0–200 ms (RTD_{0-200}), and 0–300 (RTD_{0-300}) based on a previous study (29). The onset of muscle contraction was defined as the time point at the RTD exceeding the baseline by >8 N·m/s using a custom automatized macro (Peak analyses, LabChart 8.1, ADInstruments, Australia) considering the noise of the dynamometer in the torque signal as recommended previously (24, 29).

Peripheral Nerve Stimulation

PNS was delivered using a pair of bipolar electrodes with a pulse duration of 1 ms. First, the optimal site to stimulate the brachial plexus was determined as the position where the best BB response on peak torque and sEMG maximal compound wave (M_{MAX}) was observed, together with the lowest sEMG response of TBlh at rest. This optimal site was mapped over the Erb's point using a stimulation pen (Compex, Ginebra, Switzerland). A disposable circular electrode (2.5 cm in diameter) was then fixed with tape on the identified spot (28). The optimal stimulation site was marked with indelible ink to

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ensure the electrode placement was reliable within- and between sessions. A recruitment curve of stimulation was then determined at rest. Briefly, several double stimulations were applied at increasing current intensities until there was no further increase (>2%) in peak torque and M_{MAX} amplitude. This was considered the optimal intensity (28). The stimulation intensity used for testing was 120% of the optimal intensity to ensure supramaximal stimulation (28). As shown in Fig. 1*B*, PNS was applied at rest to evoke a Tw, Db10, and Db100 with a 3-s rest between stimulations (28, 30). The peak torque in Tw, Db10, and Db100 was determined to obtain measurements of the contractile properties in the nonexercised muscle. An average of 39.1 ± 9.7 mA was delivered as 120% of the optimal intensity for PNS. The twitch interpolation technique (TIT) was used to determine the percentage of voluntary activation with PNS (%VA PNS) during active voluntary contractions (25%, 50%, 75%, and 100% of MVIC in random order). A Db100 was triggered manually at the torque plateau and was followed 5 s later by a Db100 evoked at rest (30). The percentage of voluntary activation was determined (%VA PNS) using the following formula: VA% = (1 − superimposed twitch/potentiated twitch) × 100 (30).

Transcranial Magnetic Stimulation

Single-pulse TMS was delivered at the optimal site on the primary motor cortex (M1) using a 110-mm double-cone coil. Motor-evoked potentials (MEPs) amplitude was recorded from the BB and TBlh muscles. To determine the optimal stimulation site on M1, participants wore a snug-fitted cap where nasion-inion and inter-aural lines were marked, together with a 1-cm grid in a longitude-latitude matrix (28). The TMS coil was orientated tangentially to the skull and positioned to stimulate the M1 in a posterior-anterior direction (28). The optimal site of the BB on the M1 was determined using initial exploration, during which we recorded the response to stimulation from the BB and TBlh muscles during a voluntary submaximal (20% of MVIC) contraction (31). The optimal site corresponded to the point where the largest MEP amplitude was observed, together with a lesser TBlh sEMG response based on the BB/TBlh MEP ratio. This was assessed consistently in four of six consecutive stimuli using low stimulation intensities (30%–50% of stimulator output). A TMS recruitment curve was then performed using 5% intensity increments in the stimulator output from 35% to 90% in random order, while the participant performed a voluntary contraction at 20% of MVIC (31) (Fig. 1*B*). The stimuli were delivered randomly every 3–5 s to avoid stimulus anticipation, and 30-s rest was provided between each stimulus to reduce muscle fatigue. The recruitment curve was used to determine the optimal intensity of the stimulator, which was determined as the lowest intensity evoking the highest BB MEP and lowest TBlh MEP. The stimulation intensity was set at 130% of the optimal intensity for further stimulations to ensure consistency of the maximal MEP at higher contraction intensities based on our previous pilot measurements. An average of 65.2 ± 2.8% of stimulator output (130% of optimal intensity) was used for TMS test stimulation, in the plateau of the stimulus-response curve to ensure maximal stimulation (31). For the selected TMS stimulation intensity, the ratio in the MEPs amplitude between BB/TBlh was on average 23.5 ± 24.8, resulting in a BB MEPs amplitude on average of 78.5 ± 38.8% of BB M_{MAX}. The percentage of voluntary activation using TMS (%VA TMS) was determined with the TIT, using the extrapolation of resting twitch from the linear regression between the superimposed twitch and voluntary force at different force levels (32). We used 25%, 50%, 75%, and 100% of MVIC in random order, separated by 1-min rest between contractions to assess the %VA TMS. The MEPs amplitude and cortical silent period (CSP) durations were also assessed during these contractions. The MEP amplitude was calculated as the peak-to-peak amplitude of the sEMG response of BB and was expressed as a percentage of BB M_{MAX} amplitude to consider modifications in muscle excitability. The CSP was measured manually from the end of the MEP to the resumption of sustained sEMG activity by visual inspection (33). The CSP was measured to gauge changes in corticospinal inhibitory mechanisms (33).

Data Analyses

All analyses were performed in LabChart 8.4 (ADInstruments, Australia) using custom-made semiautomatized macro analyses scripts. For the TMS- and PNS-evoked potentials, the filtered sEMG amplitude was analyzed to calculate the peak-to-peak amplitude. During the resistance exercise, the total exercise volume was calculated by multiplying the number of repetitions by the weight (kg) performed for each participant during CONC and ECC. In addition, during the exercise sessions, the sEMG amplitude was computed with the root mean square (RMS) calculation using a moving window of 10 ms, and contracting time was defined when sEMG increased above two standard deviations (SD) of baseline activity to calculate the integral of muscle activity ($sEMG_i$) (4, 28) during the MVIC and the dynamic contractions involved in the resistance exercise. Finally, the muscle activity of both BB during resistance exercise was expressed as a percentage of the highest amplitude $sEMG_i$ during MVIC. For the neuromuscular measurements analysis, the average of the two attempts for all measurements was used for statistical analyses.

Statistical Analysis

All data were first screened for normality using a Shapiro–Wilk test. Then, a one-way analysis of variance (ANOVA) was used to compare the normalized changes for each variable over time (% of Pre) between control and exercise conditions. Post hoc tests were corrected for multiple comparisons using Dunnett test corrections. When the absence of normal distributions was found, a nonparametric analysis of variance with Dunn's multiple comparisons test was performed. An unpaired (parametric or nonparametric based on normality test) *t* test was used to compare the % of change in MVIC and sEMGi in the exercised and nonexercised elbow flexors between CONC and ECC conditions. The association between the changes (% of pre) in force production (MVIC and RTD), CSE (i.e., MEPs amplitude and CSP duration), and %VA (i.e., evoked by PNS and TMS) were performed using the Pearson correlation coefficient. In addition, the effect sizes (ES) were

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calculated for the changes over time for each dependent variable in the CONC and ECC exercise conditions, being interpreted based on previous recommendations (34). The statistical analyses were performed using Prism 10.0 (GraphPad, CA) and SPSS v21.0 (IBM, NY). The statistical significance was set at $P \leq 0.05$. Descriptive data are presented as mean and standard deviation (means ± SD).

RESULTS

Baseline Values

Age ($P = 0.54$), height ($P = 0.98$), and weight ($P = 0.66$) were similar between ECC and CONC groups at baseline. As expected, the 1-RM_{ECC} (19.55 ± 4.12 kg) was 26.1% greater than 1-RM_{CONC} (15.50 ± 3.81 kg) ($P = 0.009$). No differences were found in any dependent variables at baseline between CONC and ECC, as shown in Table 1.

Fatigability of the Neuromuscular Protocol

No changes were observed in the MVIC measured at the beginning (MVIC-1) and at the end (MVIC-2) of the neuromuscular testing protocol during all conditions, showing that no fatigue was induced by the testing protocol. Values of MVIC during each condition and time are shown in Supplemental Material.

Resistance Exercise in the Dominant Elbow Flexors

As shown in Fig. 2*A*, the total exercise volume was 26.2% greater ($P = 0.010$), whereas BB $sEMG_i$ during exercise was 63.1% lesser (Fig. 2*B*) in ECC compared with CONC ($P = 0.040$). The total NASA-TLX (861.3 vs. 833.0 points, $P = 0.81$; Fig. 2*C*) and each component; mental demand (125.5 vs. 67.0 points, $P = 0.11$), physical demand (193.1 vs. 141.7 points, $P = 0.38$), temporal demand (106.6 vs. 119.3 points, $P = 0.71$), frustration (37.5 vs. 46.6 points, $P = 0.23$), effort (121.8 vs. 82.6 points, $P = 0.70$), and performance (41.8 vs. 34.1 points, $P = 0.42$) were similar between ECC and CONC. However, MVIC decreased more ($P = 0.030$) immediately after ECC (−34.3%; $P < 0.001$) than CONC (−21.7%; $P < 0.001$) (Fig. 2*D*), without changes in the sEMGi during MVIC in neither ECC (−6.0%, $P = 0.61$) or CONC (−3.1%, $P = 0.25$; Fig. 2*E*). Thus, ECC produced greater neuromuscular fatigue in the exercised limb with similar physical and mental demands compared with CONC.

Acute Contralateral Responses in the Nonexercised Elbow Flexors during Exercise

During exercise, the cross-irradiated $sEMG_i$ in the nonexercised BB was threefold greater during ECC than CONC ($P = 0.049$; Fig. 3*A*), evidencing greater motor irradiation occurred during ECC compared with CONC for the nonexercised elbow flexors.

Acute Contralateral Responses after Exercise in the Nonexercised Elbow Flexors

Force production.

Similar percentages of change were observed in MVIC torque (Fig. 3*B*) and sEMGi during MVIC (Fig. 3*C*) between all conditions.

RTD_{0-50} decreased after both ECC (−29.35 ± 23.67%) and CONC (−10.78 ± 34.57%) compared with each corresponding control condition ($CTRL_{ECC}$: 5.75 ± 16.29%, $P = 0.001$ and $CTRL_{CONC}$: 12.24 ± 13.30%, $P = 0.024$; Fig. 3*D*). However, no differences between ECC versus CONC were observed ($P = 0.33$). Slower RTD_{0-100} was observed after ECC (−23.47 ± 22.08%) compared with $CTRL_{ECC}$ (1.55 ± 15.90%, $P = 0.009$; Fig. 3*E*), whereas no differences were found between CONC compared with $CTRL_{CONC}$ ($P = 0.24$), and ECC versus CONC ($P = 0.89$). The RTD_{0-200} and RTD_{0-300} were similar between ECC versus CONC ($P > 0.99$; $P = 0.99$), ECC versus $CTRL_{ECC}$ ($P = 0.05$; $P = 0.28$), and CONC versus $CTRL_{CONC}$ ($P = 0.23$; $P = 0.85$) (Fig. 3, *F* and *G*, respectively). MVIC remained unchanged as expected, whereas the early phases of RTD may have evidenced reductions in the net neural drive to the skeletal muscle, particularly after ECC compared with CONC. No alterations occurred at peripheral levels, as indicated by the absence of changes in the later phases of RTD.

Peripheral muscular properties.

No differences in the percentage of change in peak torque of Tw, Db10, Db100, and M_{MAX} amplitude were observed between all conditions (Table 1). Thus, the muscle contractile properties remained unchanged regardless of the exercise condition.

TMS recruitment curve.

No differences in the percentage of change for optimal and supramaximal intensities ($P > 0.05$) were observed between all conditions based on TMS recruitment curve. The TMS recruitment curve MEP/M_{MAX} values from 35% to 90% of stimulator output are shown in the Supplemental Material. No global shifts in CSE across a range of stimulation intensities occurred in response to ECC and CONC in the nonexercised BB.

MEP/M_{max}.

No differences were found in the percentage of change in MEP/M_{MAX} at 25% (Fig. 4*A*), 50% (Fig. 4*B*), and 100% (Fig. 4*D*) of MVIC (Table 1). However, ECC revealed a lower MEP/M_{MAX} at 75% of MVIC (−17.0 ± 22.70%; Fig. 4*C*) compared with $CTRL_{ECC}$ (+4.63 ± 13.64%, $P = 0.02$), whereas no differences were found between CONC versus $CTRL_{CONC}$ ($P = 0.99$) and ECC versus CONC ($P = 0.10$). Thus, this shows contraction intensity-specific reduction in CSE only after ECC in the nonexercised limb.

Cortical silent period.

No differences in the percentage of change in CSP at 25%, 50%, 75%, and 100% of MVIC were found between all conditions (Fig. 4, *E–H* and Table 1). Thus, the corticospinal inhibitory processes were not altered by either ECC or CONC in the nonexercised limb.

%VA TMS and %VA PNS.

No differences in the percentage of change in %VA TMS at 25%, 50%, 75%, and 100% of MVIC were found between all conditions (Fig. 5, *A–D* and Table 1). No differences in the percentage of change in %VA PNS at 25%, 50%, 75%, and 100% of MVIC were found between all conditions (Fig. 5, *E–H* and Table 1). The net neural drive remained stable across different contraction-intensity regardless of exercise modality.

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Table 1. *Pre- and post- (means ± SD) measurements, normalized changes (% of PRE) (means ± SD), and ANOVA results of EF MVIC, BB sEMG$_I$ during MVIC, RTD at 0–50 ms (RTD$_{50}$), 0–100 ms (RTD$_{100}$), 0–200 ms (RTD$_{200}$), and 0–300 ms (RTD$_{300}$), Tw, Db10, Db100, BB M$_{MAX}$, MEP/M$_{MAX}$ at 25%, 50%, 75%, and 100% of MVIC, CSP at 25%, 50%, 75%, and 100% of MVIC, %VA TMS at 25%, 50%, 75%, and 100% of MVIC and %VA PNS at 25%, 50%, 75%, and 100% of MVIC for CTRL$_{CONC}$, CONC, CTRL$_{ECC}$, and ECC conditions*

Variable	Condition	Pre (Means ± SD)	Post (Means ± SD)	% of PRE (Means ± SD)	Anova Results	Effect Size (95% CI)
EF MVIC torque, N·m	CTRL$_{CONC}$	51.07 ± 13.1	50.22 ± 12.2	−1.02 ± 4.99	$P = 0.75$	
	CONC	53.78 ± 13.0	53.49 ± 13.2	−0.47 ± 5.83	$F = 0.40$	−0.01 (0.73–0.70)
	CTRL$_{ECC}$	55.25 ± 13.1	54.85 ± 12.3	−0.07 ± 4.94		
	ECC	54.22 ± 11.8	53.03 ± 11.5	−2.09 ± 2.90		−0.10 (0.82–0.62)
BB sEMG$_I$ MVIC, V/s	CTRL$_{CONC}$	2.44 ± 1.0	2.29 ± 1.0	−6.03 ± 18.24	$P = 0.53$	
	CONC	2.51 ± 0.7	2.52 ± 0.7	−0.34 ± 19.61	$K = 2.18$	0.01 (0.70–73)
	CTRL$_{ECC}$	2.66 ± 1.1	2.50 ± 0.9	−0.28 ± 16.56		
	ECC	2.47 ± 1.0	2.30 ± 0.9	−4.47 ± 16.75		−0.17 (0.89–0.54)
RTD$_{0–50}$, N·m/s	CTRL$_{CONC}$	84.06 ± 50.9	93.24 ± 53.6	12.24 ± 13.30	$P < 0.001$	
	CONC	87.72 ± 50.6	80.29 ± 57.5	−10.78 ± 34.57	$F = 9.47$	−0.16 (0.88–0.56)
	CTRL$_{ECC}$	112.22 ± 73.1	114.62 ± 69.2	5.75 ± 16.29		
	ECC	93.55 ± 57.0	66.24 ± 51.9	−29.35 ± 23.67		−0.49 (1.21–0.24)
RTD$_{0–100}$, N·m/s	CTRL$_{CONC}$	175.47 ± 77.8	184.49 ± 87.9	4.00 ± 17.40	$P = 0.018$	
	CONC	182.63 ± 90.73	162.40 ± 98.5	−12.66 ± 37.07	$F = 3.79$	−0.22 (0.94–0.50)
	CTRL$_{ECC}$	239.28 ± 133.3	241.69 ± 130.0	1.55 ± 15.90		
	ECC	225.99 ± 112.6	170.08 ± 94.3	−23.47 ± 22.08		−0.52 (1.25–0.20)
RTD$_{0–200}$, N·m/s	CTRL$_{CONC}$	224.59 ± 60.8	221.04 ± 72.1	−1.81 ± 19.35	$P = 0.03$	
	CONC	244.00 ± 76.6	219.31 ± 99.6	−12.00 ± 26.50	$F = 8.88$	−0.31 (1.03–0.41)
	CTRL$_{ECC}$	277.32 ± 108.3	275.13 ± 102.9	0.62 ± 9.71		
	ECC	275.99 ± 90.1	236.35 ± 74.3	−12.69 ± 16.07		−0.47 (1.19–0.26)
RTD$_{0–300}$, N·m/s	CTRL$_{CONC}$	175.99 ± 39.7	170.88 ± 40.4	−3.21 ± 11.03	$P = 0.34$	
	CONC	203.86 ± 79.3	191.40 ± 85.5	−6.93 ± 20.06	$F = 3.39$	−0.16 (0.88–0.56)
	CTRL$_{ECC}$	230.73 ± 111.6	229.14 ± 107.5	−0.68 ± 9.82		
	ECC	231.32 ± 105.6	206.47 ± 81.9	−8.06 ± 14.00		−0.26 (0.97–0.46)
Tw, N·m	CTRL$_{CONC}$	2.79 ± 1.5	2.67 ± 1.3	−3.60 ± 8.50	$P = 0.70$	
	CONC	3.06 ± 1.8	2.89 ± 1.7	−5.35 ± 17.42	$F = 0.48$	−0.09 (0.81–0.62)
	CTRL$_{ECC}$	3.20 ± 1.7	3.08 ± 1.5	−0.59 ± 11.15		
	ECC	3.05 ± 0.9	2.82 ± 1.1	−7.87 ± 25.98		−0.22 (0.94–0.50)
Db10, N·m	CTRL$_{CONC}$	4.69 ± 2.0	4.45 ± 2.1	−6.06 ± 19.50	$P = 0.64$	
	CONC	4.94 ± 2.9	4.57 ± 3.0	−10.30 ± 21.64	$F = 0.56$	−0.12 (0.84–0.59)
	CTRL$_{ECC}$	5.28 ± 2.4	5.10 ± 2.4	−3.69 ± 8.59		
	ECC	5.50 ± 1.9	4.86 ± 1.9	−10.41 ± 23.16		−0.26 (0.98–0.46)
Db100, N·m	CTRL$_{CONC}$	4.79 ± 2.9	4.49 ± 2.8	−6.52 ± 8.91	$P = 0.56$	
	CONC	4.93 ± 3.3	4.54 ± 3.0	−4.31 ± 20.35	$F = 0.69$	−0.12 (0.84–0.60)
	CTRL$_{ECC}$	6.19 ± 3.4	5.76 ± 3.2	−6.23 ± 9.39		
	ECC	5.97 ± 3.0	5.37 ± 2.6	−7.40 ± 19.88		−0.13 (0.85–0.58)
BB M$_{MAX}$, mV	CTRL$_{CONC}$	7.07 ± 2.4	7.11 ± 2.5	0.34 ± 5.35	$P = 0.99$	
	CONC	6.42 ± 2.7	6.48 ± 2.8	0.78 ± 14.77	$F = 0.01$	0.02 (0.69–0.4)
	CTRL$_{ECC}$	6.32 ± 2.8	6.39 ± 2.9	1.04 ± 5.39		
	ECC	6.99 ± 2.9	6.97 ± 2.9	0.67 ± 16.43		−0.01 (0.72–0.71)
MEP/M$_{MAX}$ at 25% MVIC, %	CTRL$_{CONC}$	0.80 ± 0.5	0.77 ± 0.4	0.98 ± 18.64	$P = 0.99$	
	CONC	0.78 ± 0.4	0.76 ± 0.4	0.56 ± 36.98	$F = 0.02$	−0.05 (0.76–0.67)
	CTRL$_{ECC}$	0.84 ± 0.4	0.81 ± 0.3	−1.03 ± 13.56		
	ECC	0.73 ± 0.4	0.70 ± 0.3	0.73 ± 21.97		−0.08 (0.80–0.63)
MEP/M$_{MAX}$ at 50% MVIC, %	CTRL$_{CONC}$	0.83 ± 0.4	0.81 ± 0.3	1.24 ± 13.15	P= 0.86	
	CONC	0.87 ± 0.4	0.83 ± 0.3	4.88 ± 42.63	$K = 0.74$	−0.11 (0.83–0.61)
	CTRL$_{ECC}$	0.99 ± 0.4	0.97 ± 0.4	0.25 ± 9.57		
	ECC	0.85 ± 0.4	0.80 ± 0.3	5.12 ± 41.25		−0.14 (0.85–0.58)
MEP/M$_{MAX}$ at 75% MVIC, %	CTRL$_{CONC}$	0.77 ± 0.4	0.82 ± 0.4	4.51 ± 15.01	$P = 0.017$	
	CONC	0.87 ± 0.4	0.93 ± 0.6	4.24 ± 29.65	$F = 3.83$	0.11 (0.60–0.83)
	CTRL$_{ECC}$	0.95 ± 0.5	1.00 ± 0.5	4.63 ± 13.64		
	ECC	0.88 ± 0.3	0.75 ± 0.4	−17.00 ± 22.70		−0.36 (1.08–0.36)
MEP/M$_{MAX}$ at 100% MVIC, %	CTRL$_{CONC}$	0.75 ± 0.3	0.79 ± 0.3	5.36 ± 14.36	$P = 0.35$	
	CONC	0.80 ± 0.4	0.80 ± 0.5	4.95 ± 35.50	$F = 1.12$	0.00 (0.72–0.72)
	CTRL$_{ECC}$	0.81 ± 0.3	0.86 ± 0.4	2.63 ± 13.23		
	ECC	0.83 ± 0.4	0.73 ± 0.4	−8.85 ± 27.05		−0.24 (0.96–0.48)
CSP at 25% MVIC, ms	CTRL$_{CONC}$	0.24 ± 0.1	0.24 ± 0.1	0.72 ± 5.37	$P = 0.98$	
	CONC	0.22 ± 0.1	0.21 ± 0.1	−1.94 ± 10.22	$F = 0.06$	−0.10 (0.81–0.62)
	CTRL$_{ECC}$	0.23 ± 0.1	0.23 ± 0.1	−0.11 ± 5.04		
	ECC	0.21 ± 0.1	0.21 ± 0.1	1.64 ± 16.06		0.00 (0.72–0.72)
CSP at 50% MVIC, ms	CTRL$_{CONC}$	0.30 ± 0.1	0.30 ± 0.1	1.08 ± 5.45	$P = 0.84$	
	CONC	0.26 ± 0.1	0.27 ± 0.1	4.32 ± 15.40	$F = 0.29$	0–10 (0.62–0.81)
	CTRL$_{ECC}$	0.29 ± 0.1	0.29 ± 0.1	1.11 ± 6.34		
	ECC	0.26 ± 0.1	0.26 ± 0.1	1.06 ± 15.15		0.00 (0.72–0.72)

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Table 1.— *Continued*

Variable	Condition	Pre (Means ± SD)	Post (Means ± SD)	% of PRE (Means ± SD)	Anova Results	Effect Size (95% CI)
CSP at 75% MVIC, ms	$CTRL_{CONC}$	0.35 ± 0.1	0.35 ± 0.1	0.55 ± 4.29	$P = 0.72$	
	CONC	0.29 ± 0.1	0.29 ± 0.1	0.01 ± 19.76	$F = 0.44$	0.00 (0.72–0.72)
	$CTRL_{ECC}$	0.34 ± 0.1	0.34 ± 0.1	0.91 ± 4.03		
	ECC	0.30 ± 0.1	0.29 ± 0.1	−0.62 ± 10.97		−0.10 (0.81–0.62)
CSP at 100% MVIC, ms	$CTRL_{CONC}$	0.34 ± 0.1	0.34 ± 0.1	0.18 ± 4.62	$P = 0.58$	
	CONC	0.32 ± 0.1	0.31 ± 0.1	−4.08 ± 12.69	$F = 0.66$	−0.10 (0.81–0.62)
	$CTRL_{ECC}$	0.34 ± 0.1	0.34 ± 0.1	−0.43 ± 5.34		
	ECC	0.30 ± 0.1	0.31 ± 0.1	2.69 ± 16.70		0.10 (0.62–0.81)
%VA TMS at 25% MVIC, %	$CTRL_{CONC}$	23.27 ± 8.9	23.81 ± 10.0	1.47 ± 8.42	$P = 0.10$	
	CONC	22.88 ± 7.7	19.60 ± 9.2	−9.44 ± 41.21	$K = 6.32$	−0.38 (1.10–0.35)
	$CTRL_{ECC}$	25.72 ± 7.7	26.49 ± 8.5	1.54 ± 8.00		
	ECC	24.95 ± 6.7	22.10 ± 5.5	−8.00 ± 25.95		−0.45 (1.18–0.27)
%VA TMS at 50% MVIC, %	$CTRL_{CONC}$	52.04 ± 18.4	52.10 ± 20.6	1.26 ± 9.78	$P = 0.24$	
	CONC	54.20 ± 15.3	59.13 ± 17.9	9.95 ± 28.53	$F = 1.43$	0.29 (0.43–1.01)
	$CTRL_{ECC}$	46.41 ± 15.5	46.66 ± 17.2	0.94 ± 10.07		
	ECC	49.66 ± 13.7	54.78 ± 11.8	19.13 ± 45.89		0.39 (0.33–1.11)
%VA TMS at 75% MVIC, %	$CTRL_{CONC}$	89.35 ± 10.3	88.26 ± 8.8	−0.16 ± 8.69	$P = 0.48$	
	CONC	87.68 ± 14.4	88.33 ± 8.8	3.11 ± 16.65	$F = 0.83$	0.05 (0.66–0.77)
	$CTRL_{ECC}$	85.60 ± 9.4	85.22 ± 7.9	0.51 ± 9.56		
	ECC	85.10 ± 10.0	89.64 ± 8.8	6.64 ± 15.72		0.47 (0.26–1.19)
%VA TMS at 100% MVIC, %	$CTRL_{CONC}$	99.84 ± 3.1	98.02 ± 4.7	0.70 ± 2.01	$P = 0.48$	
	CONC	99.95 ± 0.2	99.87 ± 0.5	−0.08 ± 0.52	$K = 2.46$	−0.20 (0.92–0.51)
	$CTRL_{ECC}$	99.03 ± 2.7	98.41 ± 4.1	−0.65 ± 2.02		
	ECC	99.55 ± 1.2	99.25 ± 2.3	−0.29 ± 2.52		−0.16 (0.88–0.56)
%VA PNS at 25% MVIC, %	$CTRL_{CONC}$	41.20 ± 17.4	41.90 ± 17.9	0.66 ± 6.73	$P = 0.10$	
	CONC	46.59 ± 22.0	49.60 ± 22.1	12.00 ± 35.89	$F = 2.27$	0.13 (0.58–0.85)
	$CTRL_{ECC}$	41.48 ± 19.7	41.21 ± 19.9	−0.57 ± 6.65		
	ECC	39.75 ± 17.3	45.20 ± 21.7	18.69 ± 29.76		0.27 (0.45–0.99)
%VA PNS at 50% MVIC, %	$CTRL_{CONC}$	92.21 ± 8.0	90.96 ± 7.9	−1.04 ± 3.79	$P = 0.07$	
	CONC	91.91 ± 8.2	93.06 ± 8.2	1.55 ± 8.18	$F = 2.64$	0.14 (0.58–0.85)
	$CTRL_{ECC}$	90.39 ± 7.6	89.94 ± 7.2	−0.55 ± 4.05		
	ECC	80.15 ± 18.1	84.90 ± 17.8	7.17 ± 14.09		0.26 (0.46–0.98)
%VA PNS at 75% MVIC, %	$CTRL_{CONC}$	99.74 ± 0.6	99.45 ± 1.8	−0.23 ± 1.76	$P = 0.56$	
	CONC	99.80 ± 0.5	99.96 ± 0.1	0.16 ± 0.54	$K = 2.08$	0.43 (0.29–1.16)
	$CTRL_{ECC}$	99.52 ± 0.7	99.15 ± 1.8	−0.44 ± 1.83		
	ECC	97.71 ± 2.9	98.37 ± 3.84	0.71 ± 3.80		0.19 (0.53–0.91)
%VA PNS at 100% MVIC, %	$CTRL_{CONC}$	99.95 ± 0.2	100.00 ± 0.0	0.04 ± 0.15	P= 0.56	
	CONC	99.93 ± 0.3	99.93 ± 0.2	0.09 ± 0.20	$K = 2.06$	0.00 (0.72–0.72)
	$CTRL_{ECC}$	99.91 ± 0.2	100.00 ± 0.1	0.05 ± 0.16		
	ECC	98.81 ± 0.7	99.96 ± 0.2	0.15 ± 0.77		2.17 (1.27–3.08)

No significant differences ($P > 0.05$) in all dependent variables were found for the PRE values between $CTRL_{CONC}$, $CTRL_{ECC}$, CONC, and ECC conditions. The effect size (ES) with 95% of confidence intervals (95% CI) are shown for the Pre- and Post-changes after CONC and ECC. ANOVA, analysis of variance; BB, biceps brachii; CONC, concentric; CSP, cortical silent period; $CTRL_{CONC}$, control concentric; $CTRL_{ECC}$, control eccentric; Db10, doublet 10 Hz; Db100, doublet 100 Hz; ECC, eccentric; EF, elbow flexors; MEP, motor evoked potentials; MEP/M_{MAX}, MEP normalized to M_{MAX}; M_{MAX}, maximal compound wave; MVIC, maximal voluntary isometric contraction; RTD, rate of torque development; $sEMG_I$, surface electromyography integral; Tw, twich; %VA PNS, voluntary activation percentage by peripheral nerve stimulation; %VA TMS, voluntary activation percentage by transcranial magnetic stimulation.

Associations between the changes in CSE and RTD. Significant correlations were found between the percentage of change of MEP/M_{MAX} at 75% of MVIC with the $RTD_{0–50}$ ($r = 0.56$; $P = 0.02$), $RTD_{0–100}$ ($r = 0.51$; $P = 0.03$), and $RTD_{0–200}$ ($r = 0.45$; $P = 0.05$) in ECC (Fig. 6, *A*–*C*). No significant association was found between the percentage of change of MEP/M_{MAX} at 75% of MVIC with the $RTD_{0–300}$ ($r = 0.39$; $P = 0.08$; Fig. 6*D*) in ECC. No significant associations were found between the percentage of change of MEP/M_{MAX} at 25%, 50%, and 100% of MVIC with $RTD_{0–50}$ ($P > 0.05$), $RTD_{0–100}$ ($P > 0.05$), $RTD_{0–200}$ ($P > 0.05$), and $RTD_{0–300}$ ($P > 0.05$) in ECC. In CONC, no significant associations were observed between the percentage of change of MEP/M_{MAX} at 25%, 50%, 75%, and 100% of MVIC with the $RTD_{0–50}$ ($P > 0.05$), $RTD_{0–100}$ ($P > 0.05$), $RTD_{0–200}$ ($P > 0.05$), and $RTD_{0–300}$ ($P > 0.05$). No significant associations were found between the percentage of change of MEP/M_{MAX} and voluntary activation (%VA TMS and %VA PNS) in all levels of MVIC in ECC ($P = 0.17–0.98$) and CONC ($P = 0.19–0.94$). The significant associations between reductions in CSE and the early phases of RTD in the nonexercised elbow flexors may suggest that the acute decrease in neural output could be a central mechanism underlying the CE induced by ECC.

DISCUSSION

The present study aimed to investigate the acute CE of a unilateral concentric (CONC) and eccentric (ECC) resistance exercise bout of the elbow flexors on force production, CSE, VA, and peripheral responses in the nonexercised contralateral elbow flexors in healthy individuals. The main findings in the contralateral nonexercised elbow flexors were: *1*) ECC and CONC did not change MVIC and BB $sEMG_i$ immediately after exercise, but ECC showed larger decreases in $RTD_{0–50}$

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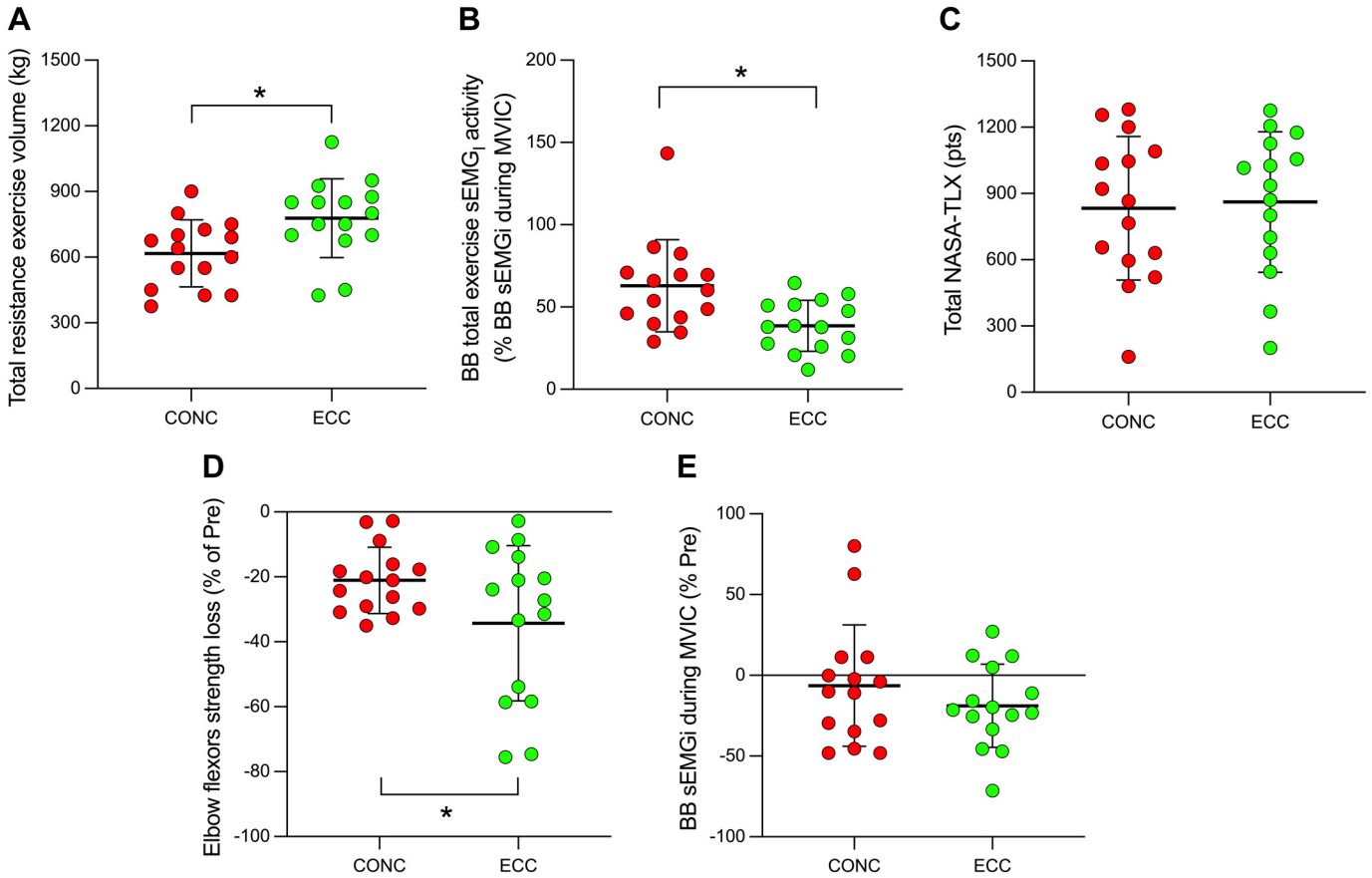

Figure 2. Total resistance exercise volume (*A*) biceps brachii (BB) total surface electromyographic integral (sEMGi) activity (*B*) during the exercise for concentric (CONC) and eccentric (ECC) conditions. Changes immediately after CONC and ECC resistance exercise for total points in NASA task load index (TLX) (*C*). Changes in the exercised elbow flexors immediately after CONC and ECC for maximal voluntary isometric contraction (MVIC) torque (*D*) and BB sEMGi during MVIC (*E*). $*P < 0.05$.

and $RTD_{0–100}$, whereas CONC only showed a reduction in the $RTD_{0–50}$ (Fig. 3 and Table 1) compared with the control conditions, respectively. *2*) Only ECC showed a decrease in CSE assessed at 75% of MVIC compared with control condition, whereas no changes were evident after CONC (Fig. 4*C*). *3*) ECC and CONC maintained the %VA measured by TMS and PNS (Fig. 5), and also maintained the peripheral factors. Finally, *4*) changes in CSE at 75% of MVIC were associated with the changes in RTD only after ECC (Fig. 6). Our results suggest that acute bouts of unilateral ECC and CONC resistance exercises modulate the neuromuscular function differently, at least in CSE and RTD in the nonexercised elbow flexors. Consequently, our preliminary hypothesis was partially supported.

Responses of the Exercised Elbow Flexors to the Unilateral Resistance Exercise

The elbow flexors $1\text{-}RM_{ECC}$ was greater than $1\text{-}RM_{CONC}$, which translated into a greater absolute training volume in ECC, similar to previously reported eccentric exercise protocols (15, 28). Despite the greater absolute training volume in ECC, it evoked lesser (−63.1%) muscle activity during exercise in the BB than CONC (Fig. 2*B*). Typically, concentric muscle contractions involve more motor unit activity compared with eccentric contractions (35). Also, greater neural inhibition and amplitude cancelation are evoked during eccentric than concentric contractions (7). Considering the greater decreases in MVIC observed after ECC (−34.3%) compared with CONC (−21.7%), these results are concordant with a previous report (36). As ECC performed resistance exercise at a greater absolute load than CONC, greater central fatigue, and muscle disruptions could be expected after ECC (28). Although ECC elicited a pronounced decrease in MVIC compared with CONC, the BB muscle activity during MVIC remained similar between both groups (Fig. 2*E*). This paradoxical finding may suggest complex mechanisms at central and peripheral levels of the neuromuscular system mediating the relationship between force production and muscle activity during MVIC after ECC (8). Therefore, the dissociation between torque loss and muscle activation suggests that distinct mechanisms underlie the torque reductions observed during maximal efforts following ECC compared with CONC exercise. In contrast to our hypothesis, we found similar mental-physical demands in the NASA TLX questionnaire after ECC and CONC (Fig. 2*C*). Conversely, Kan et al. (37) showed that a 30-min eccentric cycling (at 20% of concentric peak torque, 60 rpm) induced greater mental-physical demand than concentric cycling at the same absolute intensity and cadence. However, our ECC protocol was

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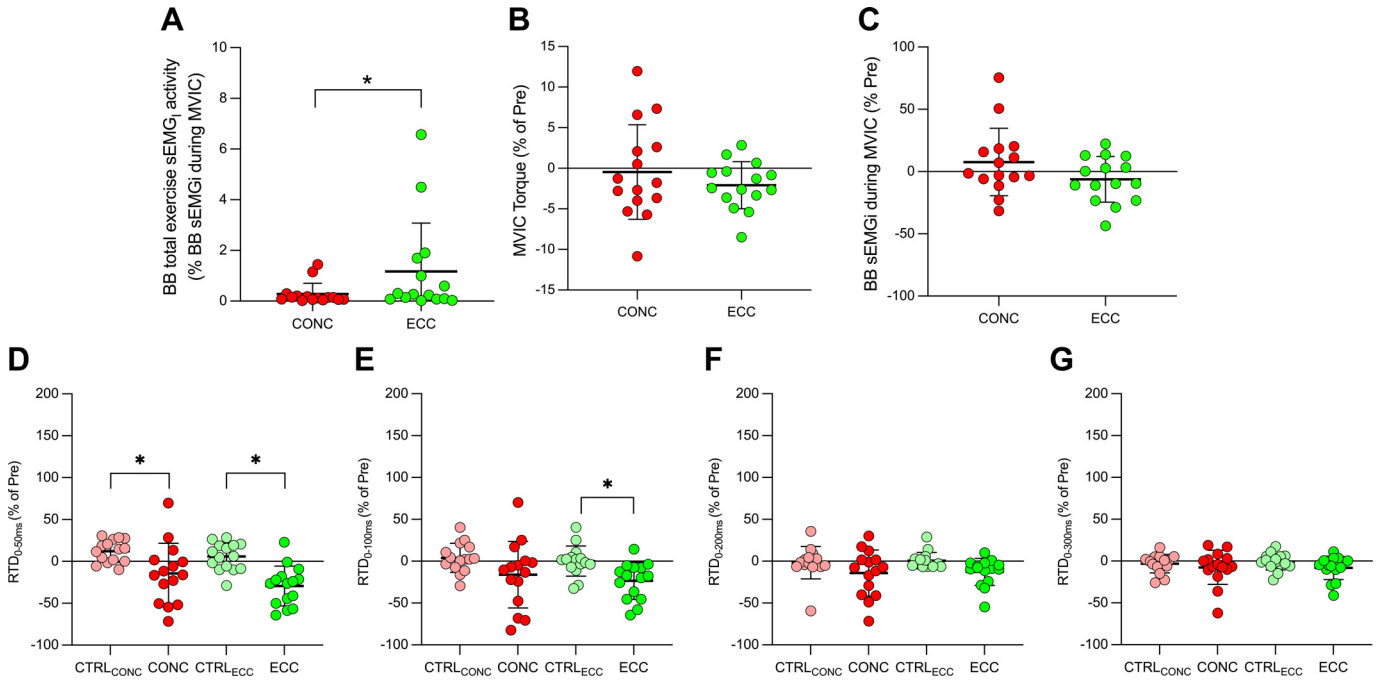

Figure 3. Biceps brachii (BB) total exercise surface electromyographic integral (sEMGi) activity in the nonexercised elbow flexors during contralateral concentric (CONC) and eccentric (ECC) exercise (*A*). Changes immediately after CONC and ECC resistance exercise on maximal voluntary isometric contraction (MVIC) torque (*B*) and BB sEMGi during MVIC (*C*) for the nonexercised elbow flexors. Changes on the nonexercised elbow flexors immediately after CONC and ECC resistance exercise for the rate of torque development in time windows from 0–50 ms ($RTD_{0–50\ ms}$) (*D*), 0–100 ms ($RTD_{0–100\ ms}$) (*E*), 0–200 ms ($RTD_{0–200\ ms}$) (*F*), and 0–300 ms ($RTD_{0–300\ ms}$) (*G*) for control concentric ($CTRL_{CONC}$), control eccentric ($CTRL_{ECC}$), CONC and ECC conditions. *Significant difference compared with each corresponding condition.

performed at the same relative intensity based on the maximal capacity for each contraction mode (1-RM_{ECC} or 1-RM_{CONC}), which may not allow individuals to perceive physical-mental demand differences at the same absolute load (37). Therefore, unilateral ECC resistance exercise seems to provide a more fatiguing stimulus for the exercised muscle than CONC at the same relative intensity, despite the similar mental-physical demands perceived by the individuals immediately after exercise.

Contralateral Effects on the Nonexercised Arm during Unilateral Resistance Exercise

The greater BB sEMG activity in the nonexercised BB during ECC compared with CONC exercise (Fig. 3*A*) could be explained by "motor irradiation" (2). The motor irradiation concept refers to the unintended motor activity of the nonexercised muscle during contralateral forceful muscle contractions (1). This might be attributed to anatomical neural connections at supraspinal and spinal levels (1, 2). Since eccentric contractions involve greater motor cortex activation (25), ECC may evoke greater widespread activity for the contralateral homologous muscles during exercise than CONC. The irradiated activity evoked by ECC could serve as a stimulus to preserve the neuromuscular function in the nontrained limb during conditions in which the direct voluntary activation of a muscle is reduced (1). However, these results related to "motor irradiation" should be interpreted with caution considering the high individual variability in the sEMG responses, particularly within the ECC group.

Contralateral Effects on the Nonexercised Arm after Unilateral Resistance Exercise

Force production changes.

ECC and CONC decreased $RTD_{0–50}$ compared with their control conditions (Fig. 3*D*), but there is a tendency for ECC (ES: −0.49) to show a larger effect in $RTD_{0–50}$ compared with CONC (ES: −0.16) based on effect size calculations. In addition, ECC decreased $RTD_{0–100}$ compared with $CTRL_{ECC}$ (Fig. 3*E*), whereas $RTD_{0–100}$ after CONC remained unchanged. It has been suggested that in acute conditions, the decreases in the early phases of RTD could represent reductions of central drive to motor neurons (23). In broad terms, early stages of RTD depend on the neural drive to facilitate the motor unit recruitment and discharge rate of active motor units, whereas late phases depend more on contractile muscle characteristics (23, 24). No changes were observed in $RTD_{0–200}$ and $RTD_{0–300}$ after ECC and CONC. Therefore, even though MVIC remained unchanged in the nonexercised elbow flexors after unilateral ECC and CONC (Table 1), our results support that early phases of RTD would be a more sensible marker compared with MVIC capacity to detect contralateral reductions in the neural drive of the nonexercised muscles after unilateral resistance exercise.

Central responses.

Despite no changes in TMS recruitment curves were evident after ECC and CONC (Supplemental Material), only ECC experienced a decrease in CSE compared with the control condition in the MEP/M_{MAX} at 75% of MVIC (Fig. 4*C*). Surprisingly, %VA TMS and %VA PNS remained unchanged

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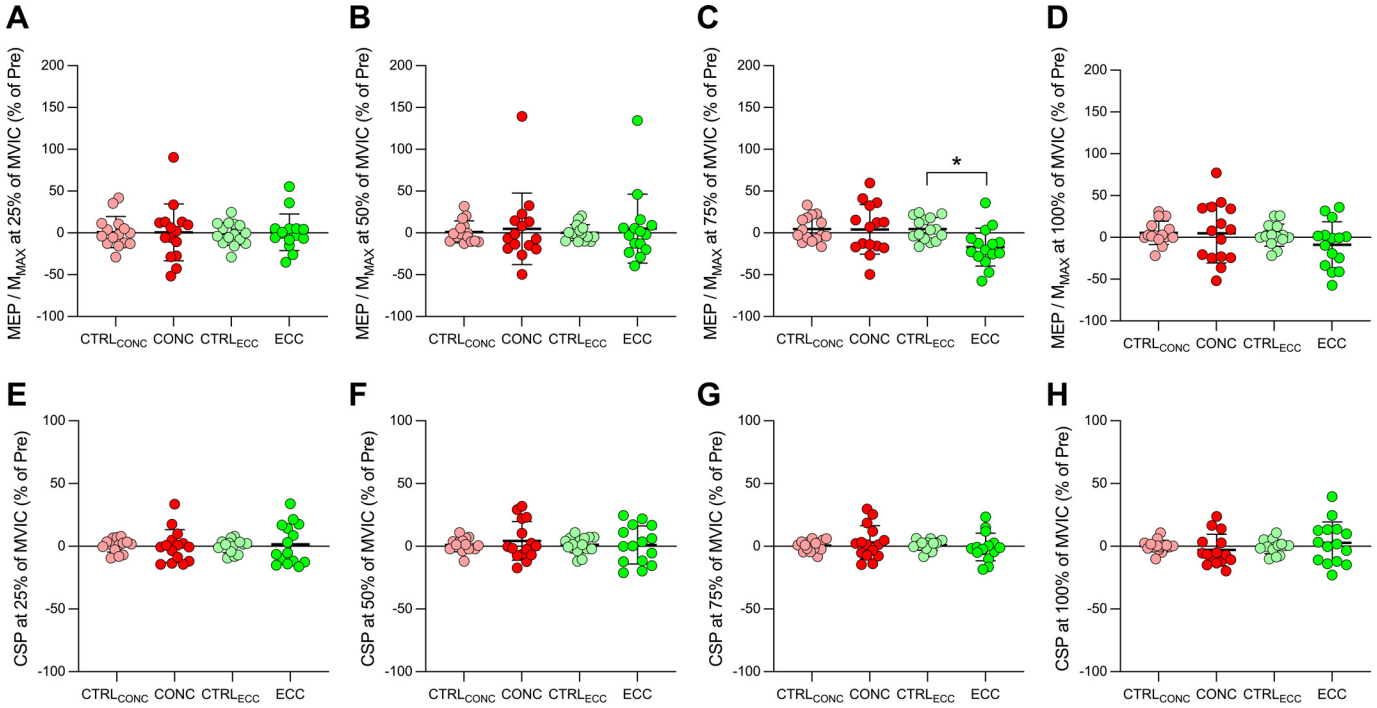

Figure 4. Normalized changes (% of PRE) on the nonexercised elbow flexors immediately after concentric (CONC) and eccentric (ECC) resistance exercise for the normalized motor evoked potentials (MEP/M_{MAX}) obtained at 25% (*A*), 50% (*B*), 75% (*C*), and 100% (*D*) of the maximal voluntary isometric contraction (MVIC) for control concentric ($CTRL_{CONC}$), control eccentric ($CTRL_{ECC}$), CONC and ECC conditions. Normalized changes (% of PRE) on the nonexercised elbow flexors immediately after CONC and ECC resistance exercise for the cortical silent periods (CSP) obtained at 25% (*E*), 50% (*F*), 75% (*G*), and 100% (*H*) of the MVIC for $CTRL_{CONC}$, $CTRL_{ECC}$, CONC, and ECC conditions. *Significant difference compared to each corresponding condition.

for both ECC and CONC (Fig. 5). In contrast to our findings, van der Groen et al. (15) did not report changes in the MEP amplitude measured at rest in the nonexercised elbow flexors after three sets of 10 repetitions at 75% of concentric and eccentric 1-RM. Unfortunately, no direct comparison can be made between the study of van der Groen et al. (15) and the present study, as CSE is intrinsically linked to specific tasks and voluntary contraction intensity (38). Our resistance exercise protocol was also performed at greater exercise intensity (80% vs. 75% of specific 1-RM) and more sets (5 vs. 3) than the mentioned study (15). Thus, it is possible that our eccentric exercise protocol was able to decrease the CSE in the nonexercised muscle due to the higher intensity and sets used compared with the previous protocol of van der Groen (15). We only found reductions in MEP/M_{MAX} at 75% of MVIC after ECC compared with $CTRL_{ECC}$, but not at the other contractions levels (25%, 50%, and 100% of MVIC). This finding aligns with prior evidence suggesting cortical and motor neuron responsiveness exhibit task-dependent characteristics. Specifically, the biceps brachii have demonstrated optimal CSE during 75% of MVIC, likely due to the balance between excitatory drive and inhibitory mechanism (32, 39). For example, stronger contractions (i.e., 90%–100% of MVIC) may lead to a ceiling effect when motorneuron recruitment plateaus, whereas weaker contraction intensities (i.e., 25%–50% of MVIC) might not provide a sufficient drive to fully engage the corticospinal pathways (32, 33). Therefore, evaluating CSE at 75% of MVIC in our unilateral resistance exercise protocol could represent a more specific and sensible evaluation of acute changes induced by unilateral resistance exercise at central levels for the nonexercised muscle. In addition, the unchanged CSP (Fig. 4, *E*–*H*) may suggest that corticospinal inhibitory responses were unaltered after ECC and CONC. However, it is important to note that we adjusted the TMS stimulation intensity based on the recruitment curve for MEP assessments, and we did not perform a specific recruitment curve for CSP as recommended (40). Thus, we cannot discard the possibility that ECC and CONC could have affected corticospinal inhibitory synapses.

Peripheral responses.

We did not observe changes in the peak torque evoked in Tw, Db10, and Db100 in the nonexercised elbow flexors (Table 1). In addition, no changes in M_{MAX} responses were detected in the nonexercised BB. Our results align with previous studies that did not find changes in peripheral markers in the nonexercised muscles after unilateral exercise (21, 41, 42). However, decreases in the electrically evoked torque in the nonexercised muscles have been reported previously after unilateral resistance exercise of the knee extensors (41) and flexors (22). It has been proposed that larger lower limb muscles (i.e., knee extensors and flexors), compared with smaller upper limb muscles (i.e., elbow flexors), could be more susceptible to nonlocal fatigue, affecting the peripheral levels due to a greater spinal inhibition evoked by unilateral exercise (22, 41). However, further studies are needed to conclude better about the acute spinal and peripheral contralateral responses following unilateral resistance exercise in the upper and lower limbs.

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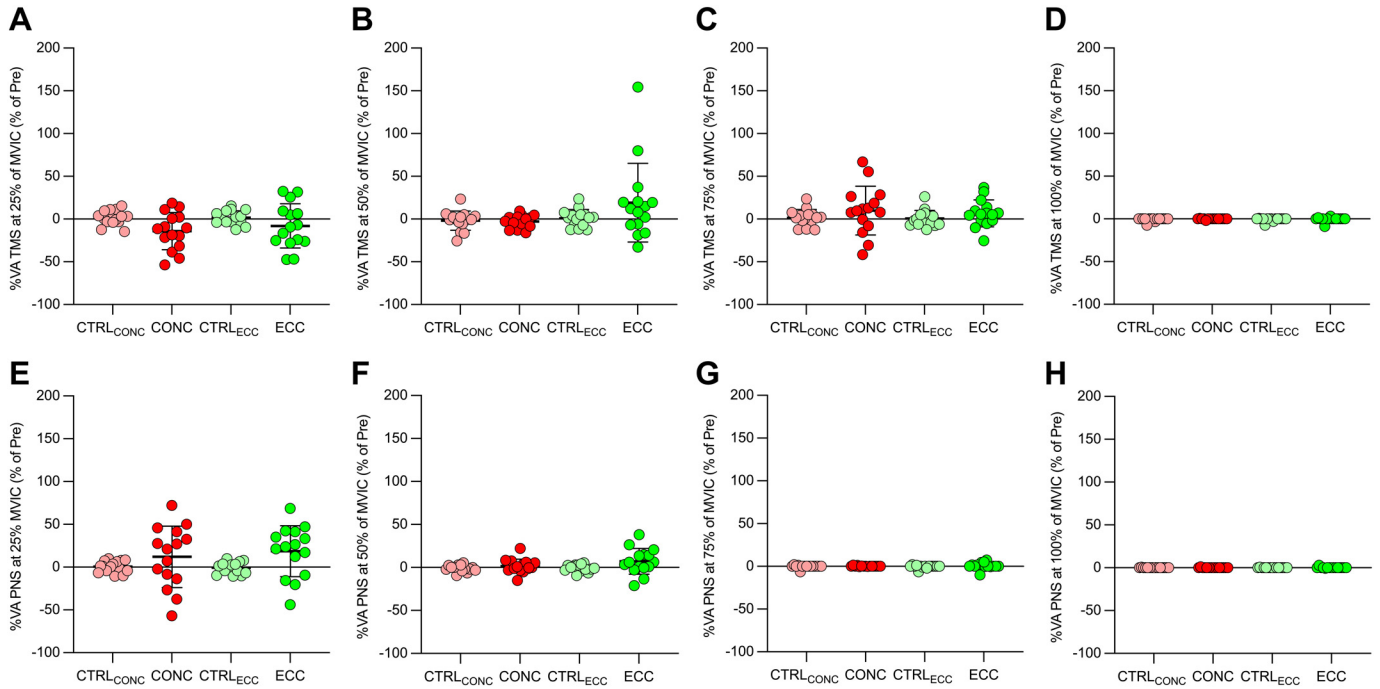

Figure 5. Normalized changes (% of PRE) on the nonexercised elbow flexors immediately after concentric (CONC) and eccentric (ECC) resistance exercise for the voluntary activation level measured with transcranial magnetic stimulation (%VA TMS) during contractions at 25% (*A*), 50% (*B*), 75% (*C*) and 100% (*D*) of the maximal voluntary isometric contraction (MVIC) concentric (CTRL$_{CONC}$), control eccentric (CTRL$_{ECC}$), CONC and ECC conditions. Normalized changes (% of PRE) on the nonexercised elbow flexors immediately after CONC and ECC resistance exercise for voluntary activation level measured with peripheral nerve stimulation (%VA PNS) during contractions at 25% (*E*), 50% (*F*), 75% (*G*), and 100% (*H*) of the MVIC for CTRL$_{CONC}$, CTRL$_{ECC}$, CONC, and ECC conditions.

Force production and the associated neuromuscular changes.

Moderate correlations between CSE and early-phase RTD (r = 0.45–0.56) in nonexercised elbow flexors were found only after ECC (Fig. 6). We speculate that the reductions in RTD and CSE observed after ECC may indicate decreased central drive, as early RTD is mainly associated to muscle activity at contraction onset (29) and changes in motor

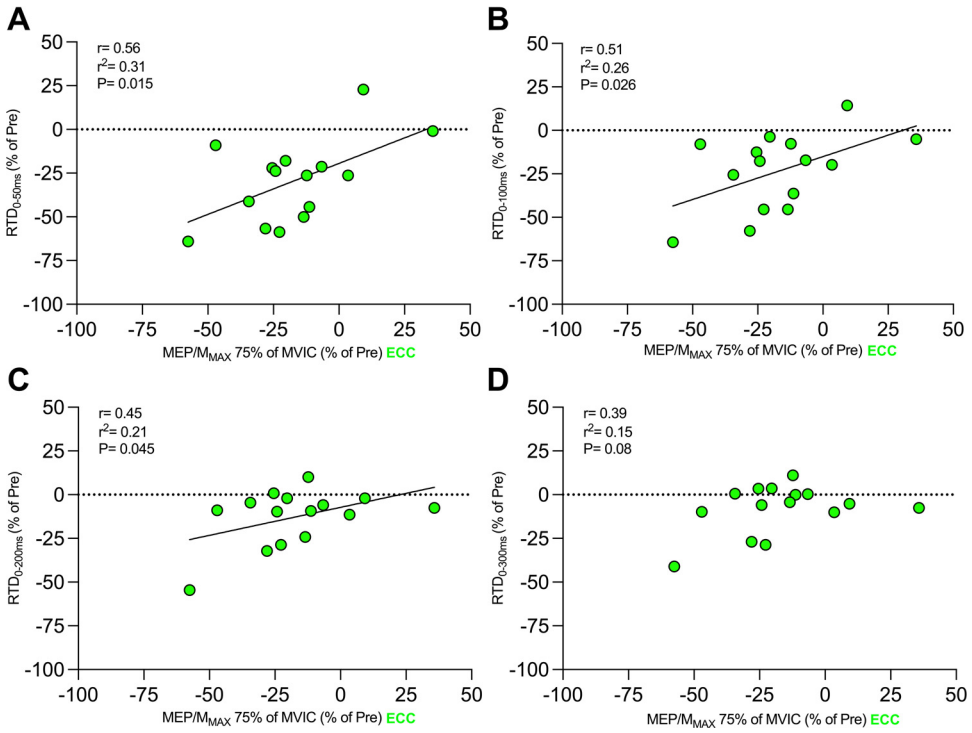

Figure 6. Correlation between the percentage of change (% of Pre) of motor evoked potential (MEP)/maximal compound wave (M_{MAX}) at 75% of maximal voluntary isometric contraction (MVIC) and the percentage of change (% of Pre) of rate of torque (RTD)$_{0–50\ ms}$ (*A*), RTD$_{0–100\ ms}$ (*B*), RTD$_{0–200\ ms}$ (*C*), and RTD$_{0–300\ ms}$ (*D*) for the eccentric (ECC) condition.

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neuron properties (23). These findings suggest that changes in CSE may partially contribute to reductions in rapid force production. However, the moderate effect sizes indicate that other factors potentially at spinal, subcortical or intracortical levels could also play a role in mediating the acute contralateral neuromuscular response. Hence, after ECC, the force production could be influenced, at least in part (r^2 = 21%–31%, Fig. 6), by the neural drive from corticospinal sources. Previous short-term training studies reported greater increases in force production in the nontrained muscles after ECC compared with CONC URT (4, 9, 12). Therefore, despite the acute reductions in RTD, we can speculate that ECC induced a more adaptative acute accumulative stimulus at central levels (42), specifically from corticospinal sources, to improve force production capacity in the nonexercised limb after short-term training compared with CONC (4, 9, 10, 12, 43). In support to this speculation, we did not find reductions in peripheral responses in the nonexercised elbow flexors after both ECC and CONC. Indeed, it is also possible that sources other than CSE, nonassessed in the present study, could have affected the net neural drive to decrease the $RTD_{0–50}$ after CONC (44). Although %VA remained unchanged after both ECC and CONC, it is important to consider that this measure whether assessed via TMS or PNS could be influenced by the neural drive originating from other central structures, which could still influence the motor neuron activity to produce a certain level of force as mentioned previously (8, 45). Thereby, further studies need to assess a broader spectrum of changes at central and peripheral levels of the neuromuscular system to draw better conclusions about the mechanism underlying the CE on force production in response to eccentric and concentric unilateral resistance exercise and short-term training.

It is also worth noting that the neuromuscular response to unilateral resistance exercise may differ depending on the muscle group studied. Some of the literature cited in the present work has investigated the CE using knee extensors (12, 17, 41), knee flexors (22), plantar flexors (21), or even elbow extensors (46). These muscles may exhibit different neural control strategies and mechanical properties compared with elbow flexors, which could influence both the magnitude and the nature of contralateral adaptations. For example, extensors present different motor unit recruitment thresholds (47) and different corticospinal excitability profiles (48) than flexors, potentially modulating the sensitivity to cross-education stimuli. Future studies are warranted to explore whether muscle group-specific characteristics affect the direction or magnitude of CE.

Implications for Further Research and Applications

Given that RFD/RTD seems to be better related to most daily tasks (49) or sports performance (50) than maximal strength or torque, further contralateral acute and short-term studies should also evaluate the associated changes in RFD/RTD. The acute and short-term changes at more comprehensive levels of the neuromuscular system need to be explored in further studies to understand better the underlying mechanism of CE on force production, which will enhance its applications for athletic training and rehabilitation programs.

Study Limitations and Further Considerations

Our study was not free of limitations. *1*) Although each exercise group (ECC and CONC) included its own control condition involving rest and repeated measurements, the addition of an independent control group could have further clarified whether the observed effects were specifically attributable to the exercise modalities. *2*) The neuromuscular measurements were performed using isometric contractions as in previous studies (9, 15, 32), and not during eccentric or concentric contractions (decreasing the intervention-measurement specificity) to avoid higher variability in the motor commands during dynamic contractions. *3*) Our testing protocol involved only immediate postmeasurements, limiting the interpretations of the time-course effect of fatigue to apply other interventions at a delayed time (i.e., 2 h or 24 h after exercise). However, the acute neuromuscular measurements performed in the present study can provide an idea of the state of the neuromuscular system to apply other interventions immediately after unilateral exercise, as occurs in real clinical setups. Finally, *4*) we only used single-pulse TMS, which only allowed us to assess CSE changes related to MEP amplitude and CSP duration. For further studies, double pulse TMS-based measures of inhibitory neurons need to be studied (e.g., intracortical inhibition and interhemispheric inhibition) to gain additional insights into the neural mechanisms underpinning changes in force production after contralateral acute resistance exercise and short-term training.

Conclusions

A single moderate-to-high intensity unilateral eccentric resistance exercise modulates the acute changes in force production and CSE of the nonexercised elbow flexors differently than concentric resistance exercise. The moderate association of the changes in RTD and CSE in the nonexercised elbow flexors after ECC may reflect a degree of reduction of the central drive to the contralateral muscles. However, it is possible that sources other than CSE also affected force production capability. These findings may suggest that ECC induces a more adaptative acute stimulus in force production and CSE than CONC unilateral resistance exercise. This could explain, at least in part, a possible underlying mechanism of the large CE by eccentric compared with concentric short-term URT (4, 9–12).

DATA AVAILABILITY

The datasets generated during and/or analyzed in the current study are available from the corresponding author upon reasonable request.

SUPPLEMENTAL MATERIAL

Supplemental Table S2: https://doi.org/10.60483/UNAB/WZCIFD/MGDQGY.

Supplemental Table S3: https://doi.org/10.60483/UNAB/WZCIFD/QD0I1G.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

O.V., E.C., J.S., S.G.-V., C.T.-J., M.C., V.M., and L.P. conceived and designed research; O.V., C.R., and O.N. performed experiments; O.V., C.R., O.N., and E.C. analyzed data; O.V., E.C., J.S., S.G.-V., C.T.-J., V.M., and L.P. interpreted results of experiments; O.V. prepared figures; O.V., C.R., E.C., J.S., S.G.-V., M.C., and V.M. drafted manuscript; O.V., O.N., E.C., J.S., S.G.-V., C.T.-J., M.C., V.M., and L.P. edited and revised manuscript; O.V., C.R., O.N., E.C., J.S., S.G.-V., C.T.-J., M.C., and V.M. approved final version of manuscript.

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